Spandan **Senapati**

CS Ph.D. student, University of Southern California

Los Angeles, California, USA

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Education

University of Southern California — *Los Angeles, California*

Ph.D. in Computer Science, advised by Prof. Haipeng Luo and Prof. Vatsal Sharan — *Aug 2023 - Present*

- Research Interests: Statistical Learning Theory, Online Learning, Trustworthy Machine Learning

Indian Institute of Technology Kanpur — *Kanpur, Uttar Pradesh*

Bachelor of Technology in Computer Science and Engineering — *July 2018 - May 2022*

- Cumulative Performance Index (CPI): **9.1/10.0** (graduated with **Distinction**)
- Awarded **Proficiency Medal** at the 55th Convocation for the best undergraduate project done by a graduating student in the CSE department
- **Academic Excellence Award** (2018-19, 2019-20): Awarded for excellent academic performance in an academic year

Experience

Apple — *Cupertino, CA*

Internship, Mentors: Parikshit Gopalan, Satyen Kale — *May 2026 - Aug 2026*

Indian Institute of Technology Kanpur — *SPiN Lab, IIT Kanpur*

Graduate Research Assistant, Advisor: Prof. Ketan Rajawat — *Aug 2022 - May 2023*

Tata Institute of Fundamental Research — *Remote*

Internship, Advisor: Prof. Rahul Vaze — *Aug 2022 - May 2023*

Microsoft — *Remote*

Data Scientist intern, Related Search Team — *May 2021 - July 2021*

Publications and Preprints

Efficient Swap Multicalibration of Elicitable Properties — *[Arxiv]*

Lunjia Hu$^{\alpha-\beta}$, Haipeng Luo$^{\alpha-\beta}$, **Spandan Senapati**$^{\alpha-\beta}$, Vatsal Sharan$^{\alpha-\beta}$ — *Nov 2025*

Annual Conference on Learning Theory (COLT), 2026

Improved Bounds for Swap Multicalibration and Swap Omniprediction — *[Arxiv]*

Haipeng Luo$^{\alpha-\beta}$, **Spandan Senapati**$^{\alpha-\beta}$, Vatsal Sharan$^{\alpha-\beta}$ — *May 2025*

Annual Conference on Neural Information Processing Systems (NeurIPS), 2025 (**Spotlight presentation**)

Simultaneous Swap Regret Minimization via KL-Calibration — *[Arxiv]*

Haipeng Luo$^{\alpha-\beta}$, **Spandan Senapati**$^{\alpha-\beta}$, Vatsal Sharan$^{\alpha-\beta}$ — *Feb 2025*

Annual Conference on Neural Information Processing Systems (NeurIPS), 2025 (**Spotlight presentation**)

Optimal Multiclass U-Calibration Error and Beyond — *[Arxiv]*

Haipeng Luo$^{\alpha-\beta}$, **Spandan Senapati**$^{\alpha-\beta}$, Vatsal Sharan$^{\alpha-\beta}$ — *Sep 2024*

Annual Conference on Neural Information Processing Systems (NeurIPS), 2024

Sharpened Lazy Incremental Quasi-Newton Method — *[PMLR][Arxiv]*

Aakash Lahoti*, **Spandan Senapati***, Ketan Rajawat, Alec Koppel — *Jan 2024*

International Conference on Artificial Intelligence and Statistics (AISTATS)

Online Convex Optimization with Switching Cost and Delayed Gradients — *[PER][PEVA][Arxiv]*

Spandan Senapati, Rahul Vaze — *Aug 2023*

IFIP Performance 2023. Full paper: Elsevier Performance Evaluation (PEVA), Extended abstract: ACM SIGMETRICS Performance Evaluation Review (PER)

Proximal Algorithms for Smoothed Online Convex Optimization with Predictions — *[IEEE Xplore][Arxiv]*

Spandan Senapati, Ashwin Shenai, Ketan Rajawat — *Aug 2023*

IEEE Transactions on Signal Processing

Note: $\alpha - \beta$ denotes alphabetical ordering, which is custom in theoretical computer science research; $*$ denotes equal contribution

Honors & Awards

2022	**Proficiency Medal**, Best undergraduate project done by a graduating student from the CSE department	*IIT Kanpur, India*
2018, 19	**Academic Excellence Award**, For excellent academic performance in an academic year	*IIT Kanpur, India*
2018	**All India Rank 161**, Joint Entrance Examination Main, among 1.5 Million Candidates	*India*
2018	**All India Rank 191**, Joint Entrance Examination Advanced, among 200000 Candidates	*India*
2017	**All India Rank 24**, KVPY Scholarship, Indian Institute of Science and Government of India	*Bangalore, India*
2016	**NTSE Scholar**, Awarded to the top 1000 students in the nation by the Government of India	*India*
2016	**IJSO OCSC**, Selected for the team selection camp of the International Junior Science Olympiad(top 35 India)	*Mumbai, India*
2017, 18	**National Top 1% ,** National Standard Examination in Physics(NSEP) among 200000 Candidates	*India*
2018	**National Top 1% ,** National Standard Examination in Chemistry(NSEC) among 200000 Candidates	*India*
2017, 18	**State Top 1% ,** National Standard Examination in Astronomy(NSEA)	*India*
2017	**3rd in State**, Represented Odisha State in the Indian National Maths Olympiad(INMO)	*India*
2016	**State Top 35**, Represented Odisha State in the Indian National Maths Olympiad (INMO)	*India*

Travel Grants

2024	**GSG Professional Development Fund**, Issued by USC
2023	**National Science Foundation (NSF) Student Travel Award**, Issued by IFIP Performance 2023

Talks

2026	**Stanford Theory Lunch**, Efficient Swap Multicalibration of Elicitable Properties	*Palo Alto*
2026	**Conference on Learning Theory (COLT)**, Efficient Swap Multicalibration of Elicitable Properties	*San Diego*
2026	**Apple Machine Learning Research**, Efficient Swap Multicalibration of Elicitable Properties	*Remote*
2026	**USC Theory Lunch**, Efficient Swap Multicalibration of Elicitable Properties	*Los Angeles*
2024	**USC Theory Lunch**, Optimal Multiclass U-Calibration Error and Beyond	*Los Angeles*
2023	**IFIP Performance 2023**, Online Convex Optimization with Switching Cost and Delayed Gradients	*Chicago*

Teaching

Spring'26	**Teaching Assistant**, Machine Learning (CSCI 567)	*USC*
Fall'24	**Teaching Assistant**, Theoretical Machine Learning (CSCI 678)	*USC*

Skills

Programming	C, C++, Python
Machine Learning Framework	Pytorch

Reviewing

Conference	NeurIPS 2024, 2025 (**Top Reviewer**), 2026 (Emergency Reviewer), ALT 2026 (Emergency Reviewer), FORC 2026 (External Reviewer), FOCS 2025 (External Reviewer), SODA 2027 (External Reviewer), ALT 2027 (Program Committee)
Journal	IEEE BITS the Information Theory Magazine

Service

SHINE Mentor

USC

USC Summer High School Intensive in Next Generation Engineering (SHINE) — *June 2024 - July 2024*

- Mentored a high school student in building an online learning-based program that can beat a human user repeatedly in a game of rock-paper-scissors

Coordinator

IIT Kanpur

Special Interest Group in Machine Learning (SIGML) — *May 2021 - Nov 2021*

- Responsible for the smooth conduction of talks in different active areas of research in Machine Learning

Student Guide

IIT Kanpur

Counselling Service — *July. 2019 - July 2020*

- Responsible for the guidance and mentorship at a personal level of 5 freshman undergraduates and the smooth conduction of various events in the Orientation Session for the incoming Batch of Y19